Quiz 8-2

Q11:

(a) $f(x) = \frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n$

$g(x) = \frac{1}{1-x^3}$ ← replace $x$ with $x^3$ on $f(x)$

$\Rightarrow g(x) = \frac{1}{1-x^3} = 1 + (x^3) + (x^3)^2 + (x^3)^3 + \cdots + (x^3)^n$
$= \sum_{n=0}^{\infty} (x^3)^n$

$\Rightarrow g(x) = 1 + x^3 + x^6 + x^9 + \cdots + x^{3n} + \cdots = \sum_{n=0}^{\infty} (x^{3n})$

$$\boxed{g(x) = \sum_{n=0}^{\infty} x^{3n}}$$

(B) $g(x) = \frac{1}{1-x^3} = \sum_{n=0}^{\infty} x^{3n}$

$\frac{d}{dx}\left((1-x^3)^{-1}\right) = (-1)(1-x^3)^{-2}(-3x^2) = \frac{3x^2}{(1-x^3)^2}$

$\left(\frac{1}{3}\right)\frac{d}{dx}\left((1-x^3)^{-1}\right) = \frac{x^2}{(1-x^3)^2} = h(x)$.

$\therefore h(x) = \frac{1}{3}\frac{d}{dx}(g(x)) = \frac{1}{3}\sum_{n=0}^{\infty}\left(\frac{d}{dx}(x^{3n})\right)$

$$h(x) = \frac{1}{3} \sum_{n=0}^{\infty} 3x^{3n-1}$$

$$\boxed{h(x) = \sum_{n=0}^{\infty} x^{3n-1}}$$

(28.

(A) $f(x) = x^2 e^{-5x}$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}$$

$e^{-5x}$ ← replace $x$ with $-5x$ on $e^x$

$$e^{-5x} = 1 + (-5x) + \frac{(-5x)^2}{2!} + \frac{(-5x)^3}{3!} + \cdots + \frac{(-5x)^n}{n!}$$

$$e^{-5x} = 1 - 5x + \frac{25x^2}{2} - \frac{125x^3}{6} + \cdots + (-1)^n \frac{x^n}{n!}$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$$

$$x^2 e^{-5x} = x^2 \left( \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!} \right)$$

$$\boxed{x^2 e^{-5x} = \sum_{n=0} (-1)^n \frac{x^{n+2}}{2!}}$$

$f(x) = \frac{1}{x}$, centered at $a = 4$

$$f(x) = f(a) + f'(a)(x-a) + f''(a)\frac{(x-a)^2}{2!} + \cdots + f^n(a)\frac{(x-a)^n}{n!} + \cdots$$

$$= \sum_{n=0}^{\infty} f^n(a)\frac{(x-a)^n}{n!}$$

$f(x) = x^{-1} = \frac{1}{x}$

$f'(x) = (-1)x^{-2} = -\frac{1}{x^2}$

$f''(x) = (-1)(-2)x^{-3} = \frac{2}{x^3}$

$f'''(x) = (-1)(-2)(-3)x^{-4} = -\frac{6}{x^4}$

$f^n(x) = (-1)^n \frac{n!}{x^{n+1}}$

when centered at $a = 4$

$= 1/4$

$= -1/16$

$= 2/64$

$= -6/256$

$= \frac{(-1)^n n!}{4^{n+1}} = \frac{(-1)^n}{4}\left[\frac{n!}{4^n}\right]$

$$f(x) = \frac{1}{4} - \frac{1}{16}(x-4) + \frac{2}{64}\frac{(x-4)^2}{2!} - \frac{6}{256}\frac{(x-4)^3}{3!} + \cdots$$

$$\cdots + (-1)^n \frac{n!}{x^{n+1}} \frac{(x-4)^n}{n!} + \cdots$$

$$f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{n!}{x^{n+1}} \frac{(x-4)^n}{n!}$$

$$\therefore \quad \boxed{f(x) = \frac{1}{x} = \sum_{n=0}^{\infty} (-1)^n \frac{(x-4)^n}{x^{n+1}}}$$